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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 09/19/2008 |
| **PI Name:** | DeLuca, Dominick Ph.D. | | |
| **Project Title:** | The Effects of Vector-Averaged Gravity on the Development of T cells | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Developmental biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology | | |
| **Space Biology Cross-Element Discipline:** | (1) Immunology | | |
| **Space Biology Special Category:** | (1) Cell Culture | | |
| **PI Email:** | deluca@u.arizona.edu | **Fax:** | FY |
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| **City:** | Tucson | **State:** | AZ |
| **Zip Code:** | 85724 | **Congressional District:** | 7 |
| **Comments:** | PI is deceased as of 10/26/2007 (info from A. Chu/ARC; 9/16/2008) | | |
| **Project Type:** | Flight,Ground | **Solicitation / Funding Source:** | 2003 Space Biology 03-OBPR-03 |
| **Start Date:** | 04/01/2004 | **End Date:** | 10/26/2007 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 2 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Grant transferred to Raphael Gruener as of 10/27/2007, when Dr. DeLuca passed away. (per A. Chu/ARC 9/2008)<br>NOTE: Received NCE to 9/30/2008 per A. Chu/ARC (6/07)<br>NCE to 9/30/2007, per A. Chu 6/23/06 | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNA04CC97G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | We have developed a microgravity organ culture dish system (MOCDS) that allows for the normal development of T cells from hematopoietic stem cell (HSC) progenitors. This system mimics the normal development of T cells outside of the body, away from stress hormones made during flight. The MOCDS can be used both on the clinostat and during spaceflight. The culture systems are easy to set up and manipulate which allows for a more detailed analysis.<br>Using this specialized system, our work has revolved around studying the effects of vector-averaged gravity on the development of T cells. In our previous work that utilized MOCDS we demonstrated the gravity-dependence of T cell development. The results from this work suggest that the full sequelae of pre-TCR signaling may not be taking place in vector-averaged gravity, as indicated by a block at the immature single positive stage of development. Since then, we have set out to determine what the underlying mechanism was for this block. Our current focus has been on analyzing the cytokine profiles for our clinorotated cultures, specifically TNF-alpha and investiagtion of the exact point of blockage using pT-alpha and proliferation studies. Using the information obtained from these and our other studies, we aim to better understand, and hopefully reverse, the effects of clinorotation.<br><br>Task Objective: Determine the mechanism by which vector-averaged gravity blocks T cell development at the ISP developmental stage using organ culture, which preserves the 3-D microenvironment of the tissue.<br><br>Task Significance: This will allow for better understanding of the mechanism underlying decreased T cell development in vector-averaged gravity. This knowledge can be applied to reversing the effects of vector-averaged gravity on developing T cells. A functional immune system is critical if humans intend to reside in environments that have minimal gravity. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Our work has shed light on the signal dampening effects of microgravity exposure. By further understanding the underlying mechanism of this microgravity-induced effect, we could initiate a new line of research centered on the treatment and prevention of other thymocyte developmental problems, such as Type I diabetes. Many T cell developmental problems involve signaling defects similar to those seen in our microgravity work. For instance, microgravity exposure led to a significant increase in the production of TNF-alpha. Other work investigating the role of TNF-alpha during T cell development has shown that over expression of this cytokine may lead to deficient negative selection and hence the production of potentially autoreactive T cells. |
| **Task Progress:** | Grant transferred to Dr. Raphael Gruener as of 10/27/2007, when Dr. DeLuca passed away (per A. Chu/ARC 9/2008). For progress and information on the project after 9/30/2007, see PI Raphael Gruener. |
| **Bibliography Type:** | Description: (Last Updated: 07/05/2019) |